NAME

mbphotomosaic − Generate a georeferenced photomosaic image from a set of navigated downlooking seafloor photographs.

VERSION

Version 5.0

SYNOPSIS

mbphotomosaic [**−−verbose −−help −−threads**=*nthreads* **−−input**=*imagelist* **−−output**=*file* **−−output−tiff −−output−png −−image−dimensions**=*width*/*height* **−−image−spacing**=*dx*/*dy*[/*units*] **−−sensor−size**=*width*/*height*[/*cellsize*] **−−fov−fudgefactor**=*factor* **−−projection**=*projection_pars* **−−altitude**=*target*/*range* | **−−standoff**=*target*/*range* **−−priority−weight**=*weight* **−−priority−heading**=*target*/*factor*/*mode* **−−trim**=*trim_pixels* **−−section**=*section_pixels* **−−section−length−max**=*value* **−−rangemax**=*range_max* **−−bounds**=*lonmin*/*lonmax*/*latmin*/*latmax* **−−bounds−buffer**=*value* **−−correction−brightness −−correction−camera−settings −−correction−range**=*target*/*coeff* **−−correction−standoff**=*target*/*coeff* **−−correction−file**=*imagecorrection.yaml* **−−reference−gain**=*gain* **−−reference−exposure**=*exposure* **−−reference−intensity**=*intensity* **−−reference−crcb**=*intensity*/*intensity* **−−dark−image−ignore**=*threshold* **−−dark−image−multiply**=*threshold*/*factor* **−−platform−file**=*platform.plf* **−−camera−sensor**=*id* **−−nav−sensor**=*id* **−−sensordepth−sensor**=*id* **−−heading−sensor**=*id* **−−altitude−sensor**=*id* **−−attitude−sensor**=*id* **−−use−left−camera** | **−−use−right−camera** | **−−use−both−cameras −−calibration−file**=*stereocalibration.yaml* **−−navigation−file**=*file* **−−tide−file**=*file* **−−image−quality−file**=*file* **−−image−quality−threshold**=*value* **−−image−quality−filter−length**=*value* **−−topography−grid**=*file*]

DESCRIPTION

Mbphotomosaic generates a single georeferenced mosaic image from a set of navigated downlooking photographs, typically taken from a camera or stereo camera rig towed or flown close to the seafloor. The program reads single or stereo pair images and their associated navigation, attitude, and camera settings from an MB-System imagelist file, projects each image's pixels onto either an assumed flat seafloor at a nominal standoff or an actual bathymetric topography grid, and writes the projected pixels into a regular output raster covering a specified geographic (or projected) area.

Each output pixel of the mosaic is written from whichever contributing input pixel is judged to have the highest priority. By default priority is based purely on how close a pixel lies to the center of its source image (pixels near the middle of a photograph are usually less distorted and less vignetted than pixels near the edges); if **−−altitude** or **−−standoff** is given, priority also favors pixels whose projected standoff distance is close to a specified target value. Priority can additionally be biased by vehicle heading using **−−priority−heading**, for example to favor images taken while running a particular compass heading. Pixels are also weeded out if their line of sight is too vertical, if the projected slant range exceeds **−−rangemax**, or if they fall in the black borders introduced by camera lens undistortion (unless a fixed **−−trim** margin is set instead).

Optional corrections can be applied to pixel intensity before it is written into the mosaic: a fixed correction for recorded camera gain and exposure settings, a simple brightness normalization, a range- or standoff-dependent correction using a supplied target/coefficient pair, or the pixel-position- and-standoff-dependent 3D correction table produced by **mbgetphotocorrection**. Reference intensity and color values, as well as handling of unusually dark images (either ignored outright or boosted by a multiplicative factor), can also be specified. Camera images can be undistorted and, for stereo pairs, restricted to the left camera, the right camera, or both, using a supplied stereo camera calibration file.

The output raster's dimensions, pixel spacing, geographic bounds, and map projection can all be controlled explicitly; if left unspecified, suitable defaults are derived from the imagelist's navigation, the target standoff, and the camera field of view. The output image is written as a GeoTiff or PNG file according to the output filename's suffix, or as forced by **−−output−tiff**/**−−output−png**. **Mbphotomosaic** can process images using multiple threads (see **−−threads**), and can incorporate an external tide correction, a per-image quality filter, and an MB-System platform offset model in the same manner as **mbgetphotocorrection**.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag causes the program to print out a description of its operation and then exit immediately.

−−verbose

The **−−verbose** option causes **mbphotomosaic** to print out status and debug messages. Using it more than once increases the verbosity level.

−−threads=*nthreads*

Sets the number of processing threads used to work through the image list. The requested number of threads is limited to the number of hardware threads available and to the MB-System maximum thread count. The default is 1 thread.

−−input=*imagelist*

Sets the filename of the input MB-System imagelist file listing the images (single or stereo pairs) to be mosaicked, along with their navigation, attitude, and camera settings. The default input filename is *imagelist.mb−1*.

−−output=*file*

Sets the filename of the output mosaic image. If the filename ends in *.png* the output format defaults to PNG; otherwise the output format defaults to GeoTiff unless overridden by **−−output−tiff** or **−−output−png**.

−−output−tiff

Forces the output mosaic image to be written in GeoTiff format.

−−output−png

Forces the output mosaic image to be written in PNG format.

−−image−dimensions=*width*/*height*

Sets the output mosaic image's dimensions in pixels. If not specified, the default is 1000 by 1000 pixels, adjusted as needed to be consistent with any pixel spacing and geographic bounds that are set.

−−image−spacing=*dx*/*dy***[**/*units***]**

Sets the output mosaic pixel spacing in the x and y directions, in the given *units* (default *meters*). Appending an exclamation point to the value (for example **−−image−spacing=0.05/0.05!**) gives the requested spacing priority over the requested image dimensions when the two are in conflict.

−−sensor−size=*width*/*height***[**/*cellsize***]**

Sets the camera sensor's physical width and height in millimeters and, optionally, its pixel cell size in millimeters, used together with the camera's field of view to project pixels onto the seafloor. The defaults are 8.789 mm by 6.610 mm with a 0.00454 mm cell size.

−−fov−fudgefactor=*factor*

Sets a multiplicative fudge factor applied to the camera field of view angles. The default value is 1.0 (no adjustment).

−−**projection**=*projection_pars*
Sets the map projection used for the output mosaic (in the form accepted by MB-System's projection parameter parsing, e.g. a UTM zone specification). If not specified, the output mosaic remains in geographic (longitude/latitude) coordinates.

−−**altitude**=*target*/*range*

−−**standoff**=*target*/*range*
These two equivalent options set a target standoff distance and an acceptable range around it, both in meters. When set, pixel priority favors pixels whose projected standoff distance is close to *target*, in addition to the default centrality-based priority. The default target standoff is 3.0 meters with a range of 1.0 meter, but this standoff-based priority contribution is only enabled once one of these options is used.

−−**priority−weight**=*weight*
Sets the relative weight given to the standoff-based priority contribution (enabled by −−**altitude**/−−**standoff**) versus the default centrality-based priority. The default is 1.0.

−−**priority−heading**=*target*/*factor*/*mode*
Sets a target vehicle heading, in degrees, and a weighting factor used to additionally favor pixels from images taken close to that heading. If *mode* is omitted or the target heading alone is given, a single-direction preference is used; supplying a nonpositive target disables heading-based priority (the default state).

−−**trim**=*trim_pixels*
Sets the number of pixels to ignore around the margins of each undistorted image. If this option is not used (the default, 0 pixels), the program instead automatically ignores any purely black pixel and any pixel adjacent to a purely black pixel, which is intended to exclude the black borders that can appear around the edges of an undistorted image.

−−**section**=*section_pixels*
Sets the size, in pixels, of image sections processed together. The default is 0.

−−**section−length−max**=*value*
Sets the maximum allowed length, in output grid cells, of the longest diagonal of an image section's projected quadrilateral in the output mosaic. Sections that project to a longer diagonal than this (for example due to an oblique viewing angle near the edge of an image) are excluded from the mosaic. The default is 0.5.

−−**rangemax**=*range_max*
Sets the maximum acceptable projected slant range, in meters, from the camera to a pixel's location on the seafloor; pixels with a greater range are excluded from the mosaic. The default is 200.0 meters.

−−**bounds**=*lonmin*/*lonmax*/*latmin*/*latmax*
Sets the geographic (or projected, if −−**projection** is used) bounds of the output mosaic explicitly, in the same bounds syntax accepted by other MB-System programs (west/east/south/north order, with a variety of coordinate notations supported).

−−**bounds−buffer**=*value*
Sets a buffer distance, in meters, added around the mosaic bounds automatically derived from the imagelist's navigation when −−**bounds** is not used explicitly. The default is 6.0 meters.

−−**correction−brightness**
Selects a correction mode that normalizes each image's average intensity (after gain/exposure correction) to a reference value.

−−**correction−camera−settings**
Selects a correction mode that normalizes pixel intensity for the camera's recorded gain and exposure settings relative to the reference gain and exposure values, without any additional brightness, range,

or standoff correction.

−−correction−range=*target/coeff*
Selects a correction mode that adjusts pixel intensity, after gain/exposure correction, based on the pixel's projected slant range from the camera, using the given target range and coefficient.

−−correction−standoff=*target/coeff*
Selects a correction mode that adjusts pixel intensity, after gain/exposure correction, based on the pixel's projected standoff distance from the seafloor, using the given target standoff and coefficient.

−−correction−file=*imagecorrection.yaml*
Selects a correction mode that uses a precomputed 3D pixel-position-and- standoff correction table, as generated by **mbgetphotocorrection**, to correct pixel intensity after gain/exposure correction. If the given filename does not already end in *.yml*, that suffix is appended. If none of the correction mode options is used, no intensity correction at all is applied to the mosaic.

−−reference−gain=*gain*
Sets the reference camera gain value, in dB, used by the gain/exposure and other correction modes. The default is 0.0.

−−reference−exposure=*exposure*
Sets the reference camera exposure time, in microseconds, used by the gain/exposure and other correction modes. The default is 0.0.

−−reference−intensity=*intensity*
Sets a positive reference Y (luminance) intensity value used by the brightness correction mode. If not set, an internal default value is used.

−−reference−crcb=*intensity/intensity*
Sets positive reference Cr and Cb (color) intensity values used together with the brightness correction mode.

−−dark−image−ignore=*threshold*
Sets a positive average-intensity threshold below which an entire image is ignored as too dark to usefully contribute to the mosaic.

−−dark−image−multiply=*threshold/factor*
Sets a positive average-intensity threshold and a positive multiplicative factor used to boost the intensity of images darker than the threshold, rather than discarding them outright.

−−platform−file=*platform.plf*
Sets the filename of an MB-System platform definition file describing the geometric offsets between the vehicle's navigation, attitude, and camera sensors.

−−camera−sensor=*camera_sensor_id*
Sets the platform sensor id number of the stereo camera used to look up the camera's position and orientation offsets within the platform model.

−−nav−sensor=*nav_sensor_id*
Overrides the platform model's navigation (position) source sensor id number.

−−sensordepth−sensor=*sensordepth_sensor_id*
Overrides the platform model's pressure/depth source sensor id number.

−−heading−sensor=*heading_sensor_id*
Overrides the platform model's heading source sensor id number.

−−altitude−sensor=*altitude_sensor_id*
Sets the platform sensor id number used as the altitude source.

−−attitude−sensor=*attitude_sensor_id*
Overrides the platform model's roll/pitch and heave source sensor id number.

−−use−left−camera
Process only the left image of each stereo pair (or single images acquired by the left camera).

−−use−right−camera
Process only the right image of each stereo pair (or single images acquired by the right camera).

−−use−both−cameras
Process both the left and right images of each stereo pair. This is the default camera usage mode.

−−calibration−file=*stereocalibration.yaml*
Sets the filename of an OpenCV YAML file containing the intrinsic and extrinsic stereo camera calibration parameters (camera matrices, distortion coefficients, and the rotation/translation between the two cameras) used to undistort images that are not already rectified.

−−navigation−file=*file*
Sets the filename of an ASCII navigation file used in place of, or to supplement, navigation carried in the imagelist file.

−−tide−file=*file*
Sets the filename of an ASCII tide file (time and tide height pairs) used to apply a tidal correction to the water depth used in the photomosaic projection.

−−image−quality−file=*file*
Sets the filename of an ASCII file of time-stamped image quality values used to filter out low quality images.

−−image−quality−threshold=*value*
Sets the minimum image quality value (in the range 0 to 1) required for an image to contribute to the mosaic.

−−image−quality−filter−length=*value*
Sets a time filter length, in seconds, applied when interpolating image quality values from the image quality file.

−−topography−grid=*file*
Sets the filename of a bathymetric grid (in a format read by **mbgrid**/GMT) used to intersect each pixel's line of sight with the actual seafloor topography rather than an assumed flat seafloor at a nominal standoff. Specifying this option enables use of the topography grid.

SEE ALSO

mbsystem(1), **mbgetphotocorrection**(1), **mbphotogrammetry**(1), **mbimagelist**(1), **mbm_makeimagelist**(1), **mbtiff2png**(1), **mbgrid**(1), **mbmosaic**(1)

BUGS

Maybe. Maybe not.